

SPURTS OF GROWTH

Here's a thought: Each time humans make a breakthrough, their rate of growth speeds up. They then take advantage of it. They fill up the economic niche that opens up for them as fully as their new technology allows. And then what? Growth then goes back to "normal." But what's normal? Apparently, normal is very low or negligible rates of growth; that's what we had before the Industrial Revolution began. For approximately 199,700 years, the growth rate of the human species was negligible. Vanishingly small. Zilch. For thousands of years at a stretch, there was no change in populations or living standards.

Which, of course, only gives rise to a whole group of questions.

First, is it true? Is human history a long spell of stagnation or low levels of growth, punctuated by sudden bursts of above-trend growth?

If it is true, is it also true that the developed economies have reached the limit of the "cheap energy" dividend, which began with the large-scale exploitation of coal in the eighteenth century, followed by the wide use of oil in the nineteenth and twentieth centuries?

And if that is true, what next? Are there no new sources of cheap energy?

What about nuclear? Well, a possibility, but recently, nuclear power was dealt a huge setback. It's not cheap, not when you add in all the safety features and the cost of the occasional emergency. Many nations are now reexamining their energy policies to decide whether nuclear has any role to play at all.

What about conservation, energy-saving measures, going green?

Well, yes, you can stretch your energy. You can probably even increase your quality of life by using it more efficiently. But you don't get above-trend economic growth gains by reducing the inputs of energy. You appear to get rapid growth only from big breakthroughs that make new energy available—and put it to work for you.

Doesn't technology lead to big increases in standards of living? To growth? To higher gross domestic product (GDP)?

Apparently not. The biggest new technology to come along in recent years is the Internet and the accompanying communications revolution. Nowhere was this more promising or more fully elaborated than in the